

# BLM1011 Introduction to Computer Science

## Assignment - I

**Due 08/11/2020 – 23:59**

**Instructor: Assist. Prof. M. Amaç GÜVENSAN**

**Question 1:** Suppose a digital camera has a storage capacity of 500MB. How many black-and-white photographs could be stored in the camera if each consisted of 512 pixels per row and 512 pixels per column if each pixel required one bit of storage?

**(6 Points)**

**Question 2:** What bit patterns are represented by the following hexadecimal notations?

**a. 8A9      b. DCB      c. A01      d. C99 (8 Points)**

**Question 3:** What is the value of the least significant bit in the bit patterns represented by the following hexadecimal notations?

**a. 9A      b. 90      c. 1B      d. 6E (8 Points)**

**Question 4:** Convert each of the following base 10 representations to its equivalent excess sixteen representation:

**a. -12      b. 0      c. 10      d. -8 (8 Points)**

**Algorithm 1:** Design an algorithm which converts the given zeros and ones into a fractional number (use base 10 representation) using the given point position? **(70 Points)**

**Input (should be given by the user) :**

**N :** the number of elements of the given array

**Array:** consists of zeros and ones

**Point:** the position of the point

Example 1:

N=6

Array= 1 1 1 0 1 0

Point = 3

Hint = 111.010

**Answer: 7.25**

Example 2:

N=5

Array= 1 0 0 1 1

Point = 2

Hint : 10.011

**Answer: 2.675**

Example 3:

N=4

Array= 1 1 0 0

Point = 1

Hint : 1.100

**Answer: 1.5**

Example 4:

N=3

Array= 0 1 0

Point = 2

Hint : 01.0

**Answer: 1**

\* Negative numbers could not be represented in this algorithm.

## SUBMISSION

1. Assignments submitted after submission deadline will not be evaluated.
2. Collaboration on any assignment is strictly prohibited. Submitted assignments are automatically checked for similarities. Infractions will be given a zero for the entire assignment.
3. Assignments **MUST** be submitted via [online.yildiz.edu.tr](http://online.yildiz.edu.tr)
4. You should submit your PDF file with the name of your studendid as given below.

Example File Name : [18011001.pdf](#)

## Content

An PDF file which contains

- a. **Answer** - should include detailed solution steps for question 1,2,3,4
- b. **Flowchart** - Flowchart for Algorithm 1. (You can draw your flowchart either by your hand or by computer. Just work clean !!! )
- d. **Analysis** - should be done for Algorithm 1 for each possible different cases

Do not forget to prepare a cover page which should include

- Course Name
- Course Group
- Instructor Name
- Assignment Number
- Delivery Date of the Assignment
- Student Id
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## ATTENTION

- **Assignments that don't comply with submission rules will NOT be evaluated. "NO EXCEPTION"**